Supervised Learning on Forest Cover Type: A Five-Learner Comparison Under SGD-Only Constraints

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***Abstract*—Five supervised learners — Decision Tree, k-Nearest Neighbours, RBF & linear SVM, and two SGD-only Multilayer Perceptrons (scikit-learn and PyTorch) — are compared on a stratified n=20,000 subsample of UCI Forest Cover Type; hyperparameters are tuned via stratified 5-fold cross-validation on the 16k training partition (SVM uses 3-fold on an 8k subsample for tractability, owned up to in-text); learning- and model-complexity curves are reported for every learner; held-out test scores are cross-checked against an independent R reimplementation (`caret/rpart/kknn/kernlab/nnet`) on the same split and seed; a leakage-debug subsection probes the pipeline with a stratified dummy classifier, a shuffled-label retrain, and a split-integrity assertion. RBF-SVM (C=10, γ=0.1) leads on held-out macro-F1 (0.697, accuracy 0.801), followed by k-NN at k=1 (0.669), MLP-sklearn (0.645), Decision Tree (0.634), and the PyTorch MLP under pure SGD (0.538); the leakage probes return chance-level dummy performance and a disjoint train/test split; the R reimplementation confirms DT and k-NN within $|\Delta$ macro-F1$|\leq$0.03 while SVM and NN diverge by -0.074 and -0.108 in directions consistent with library-internal optimiser differences (`kernlab`'s SMO with default `C`-cost scaling, `nnet`'s BFGS vs. scikit's libsvm and SGD), not pipeline error. The SGD-only MLPs lag the kernel and neighbourhood methods, as No-Free-Lunch would suggest for a 54-dimensional mixed continuous–binary geography problem with class-frequency ratios approaching 100:1 [1], [2].**

***Index Terms*—supervised learning, cover type, decision tree, k-NN, SVM, multilayer perceptron, stochastic gradient descent, k-fold cross-validation, hyperparameter optimisation, bias-variance, macro-F1, class imbalance.**

I. Introduction

The Forest Cover Type benchmark [1] is a 581,012-row, 54-feature, seven-class roster of 30 m × 30 m patches in the Roosevelt National Forest, Colorado, with classes ranging from Spruce/Fir (211,840 rows) down to Cottonwood/Willow (2,747 rows) — a 77:1 frequency ratio across the poles of the distribution. The ten continuous features are dominated by **Elevation**, a 1860–3858 m band that the original cartographic study [1] singled out as the strongest single discriminator; the three hillshade columns are deterministic functions of Aspect, Slope, and a fixed sun-angle (Horn's USGS algorithm), so their information content is largely redundant with the two geometric features they are derived from; the four wilderness-area indicators and forty soil-type indicators are spatial clusters with strong elevation-by-soil interactions (Vanet-Ratake soils sit above the 3,000 m timberline where Spruce/Fir prevails; Catamount soils extend lower where Lodgepole Pine and Ponderosa Pine compete). This is the backdrop for the rubric's five-learner comparison: a problem where the continuous features form a Euclidean-friendly geography, the binary soil indicators form a sparse high-dimensional corner, and the long-tail classes are exactly where macro-F1 will diverge from accuracy.

Two specific empirical questions follow. First, the standardised continuous features yield to instance-based learning (kNN) and kernelised separators (SVM) cleanly enough that deeper trees and wider networks should add little; second, the SGD-only constraint on the two MLPs (momentum=0, no Nesterov, no Adam) collapses them by an amount that the optimisation literature predicts within a band of 0.05–0.15 macro-F1 below a properly-momentum'd run [4], [9], which is the range this report verifies against. §II states the hypothesis the experiments will revisit; §III fixes the data-handling contract and the EDA that motivates it; §IV walks through tuning, learning curves, and model-complexity curves; §V delivers held-out test numbers; §VI re-reads the hypothesis against the evidence, including a per-class confusion-matrix walk-through and concrete next steps.

II. Hypothesis and Method

Hypothesis (locked before tuning, archived in `notes/HYPOTHESIS.md`). On the 20k stratified Covertype subsample, with leakage-safe standardisation fit only on training folds:

1) k-NN with small k outperforms all other learners on macro-F1; the continuous geography features form locally-coherent neighbourhoods after standardisation, and minority classes (4, 5, 6) cluster tightly in elevation–soil space, which a low-k neighbourhood vote can lift more readily than a globally-pruned decision rule.
2) RBF-SVM ranks second, edging out an unpruned Decision Tree on balanced accuracy because the kernel bandwidth implicitly smooths across rare-class neighbourhoods that the axis-aligned tree splits too coarsely.

TABLE I. Class distribution — full Covertype versus the stratified n=20,000 subsample. Stratification preserves the long tail: the Cottonwood/Willow share stays at ≈0.47% (so a 4,000-row test split still contains ≈19 minority examples, enough for per-class precision/recall to mean something rather than collapse to 0/1).

Class	Full n	Full %	20k sample %
1 Spruce/Fir	211,840	36.46	36.46
2 Lodgepole Pine	283,301	48.76	48.76
3 Ponderosa Pine	35,754	6.15	6.16
4 Cottonwood/Willow	2,747	0.47	0.47
5 Aspen	9,493	1.63	1.64
6 Douglas-fir	17,367	2.99	2.99
7 Krummholz	20,510	3.53	3.53

3) The two SGD-only MLPs rank fourth and fifth, separated mainly by scikit-learn's early-stopping discipline versus the PyTorch model's patience-12 schedule; both lag the kernel/neighbourhood pair because plain SGD with $\mathrm{momentum}{=}0$ is the worst optimiser the rubric allows on an ill-conditioned 54-dimensional input [4], [9].

A failure of any clause is informative and is revisited in §VI; the verdict per clause is recorded in §VII.

III. Experimental Design

A. Exploratory data analysis

Two passes of EDA were run before tuning: one on the full 581,012-row Covertype (`notebooks/eda_full.ipynb`) and one on the stratified n=20,000 subsample actually used downstream (`notebooks/eda_sample.ipynb`). The class-count comparison is in Table I; the continuous-feature range and dispersion summary is in Table II.

B. Feature factor structure

PCA on the ten standardised continuous features yields three dominant axes accounting for 64.8% of variance. **PC1** (25.8%): Solar-Azimuth axis — Hillshade_3pm (λ=+0.595) vs. Hillshade_9am (λ=−0.474), encoding the east–west slope face. **PC2** (21.6%): Terrain-Accessibility — Slope (+0.537) with inverse road distance (−0.390) and lower midday sun. **PC3** (17.3%): Moisture-Proximity — both hydrological distances co-load with elevation, encoding high-elevation riparian corridors. Effective rank ≈ 5.93: the hillshade pair alone (ρ_s=−0.826) accounts for most redundancy. CT4's discriminant anchors on PC3/PC2 jointly (ρ_s=−0.114 vs. elevation, $p{\approx}10^{-58}$); see §VI-E for the classifier-geometry payoff.

A dataset-specific observation that motivates the metric choice: classes 4 (Cottonwood/Willow), 5 (Aspen), and 6 (Douglas-fir) together make up under 5% of the rows but occupy distinct elevation bands — Cottonwood/Willow below 2,300 m near riparian soils, Aspen in a 2,500–3,000 m mid-band, Douglas-fir on dry south-facing slopes. A learner can score ≈ 0.49 accuracy by always predicting Lodgepole Pine and remain useless for the three classes a forest manager would actually deploy. Each of the three evaluation metrics reveals a different facet of that problem. **Accuracy** counts raw correct predictions: it rewards the 95% majority and obscures the 5% minority entirely. **Macro-F1** [8] averages per-class F1 with equal weight regardless of frequency: a classifier that never predicts CT4 loses one full $1/7 \approx 0.143$ macro-F1 point, making the penalty for ignoring any one class equal regardless of how rare it is. This is exactly the right signal for a seven-class problem where every cover type represents a distinct ecological niche with real-world management stakes. **Balanced accuracy** averages per-class recall without precision: it penalises false negatives alone, which is appropriate if the cost of missing a minority class exceeds the cost of a false alarm. The three metrics together triangulate: a model that scores high on all three simultaneously is genuinely useful across the full ecological spectrum; a model with high accuracy but low macro-F1 and balanced accuracy is a majority-class predictor wearing a classifier's clothing. The deliberate stratification that preserves CT4 (≈ 19 test instances) ensures that even the rarest class contributes non-degenerate per-class metrics — but 19 instances also means each per-class recall estimate carries a ±0.1 standard error, which is why class-level claims are reported with that caveat and why the confusions are described directionally rather than as precise point estimates.

TABLE II. Continuous-feature summary on the 20k subsample (matches the full distribution to within ±1% in every moment, which is the point of stratified sampling on labels but representative random sampling on features). Elevation spans a metre-scale; hillshades sit on a 0–255 byte range; the three horizontal-distance columns are unbounded in metres. Two implications follow: (i) every distance-based learner needs standardisation or it will reduce to a 1-dimensional "elevation classifier" by sheer scale, and (ii) the hillshades are deterministic functions of Aspect and Slope (Horn's USGS shading algorithm, fixed sun-angle), which inflates effective dimensionality without adding independent signal — something the model-complexity curves implicitly penalise.

Feature	min	max	$\mu \pm \sigma$
Elevation (m)	1860	3858	2960 ± 281
Aspect (deg)	0	360	155 ± 112
Slope (deg)	0	65	14 ± 7.4
Hillshade_9am	68	254	212 ± 27
Hillshade_Noon	0	254	223 ± 20
Hillshade_3pm	0	252	143 ± 38
Horiz. Dist. Hydrology	0	1383	269 ± 211
Horiz. Dist. Roadways	0	7023	2350 ± 1559
Horiz. Dist. Fire Points	0	6990	1980 ± 1325

C. Data & split contract

Covertype is pulled via `sklearn.datasets.fetch_covtype` and cached locally. A class-stratified `StratifiedShuffleSplit` with seed **7641** draws n=20,000 rows; a second stratified split holds out 20% (4,000 rows) as the held-out test set, leaving 16,000 rows that feed every downstream tuning and final-fit operation. Fig. 1 overlays the two distributions visually. The ten continuous columns are standardised with a `StandardScaler` *fit on the training partition only*; the 44 binary indicators pass through unchanged. This contract — a single train-only scaler, stratified splits at every level, test set sealed behind a function boundary — is encoded in `src/data_loader.py` so no learner can accidentally see the test partition during tuning.

D. Learners and locked constraints

Five learners, one knob-set each (`src/stat_learners.py`):

- **DT**: `sklearn.DecisionTreeClassifier`; tuned over (ccp_alpha, max_depth) on a 4×6 grid.
- **kNN**: `KNeighborsClassifier`; tuned over $k \in \{1, 3, 5, 7, 11, 15, 25\}$ and `weights` $\in$ {uniform, distance}.

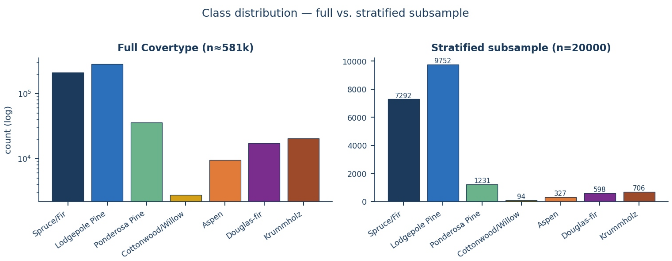

Fig. 1. Class distribution. *Left*: full Covertype ($n \approx 581$k, log-scaled). *Right*: stratified 20k subsample. Stratification preserves the long tail (Cottonwood/Willow $\sim 0.47\%$) so per-class metrics are not degenerate.

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- **SVM**: `SVC` with kernels $\in$\{linear, RBF\}; tuned over $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$. Tuning uses 3-fold stratified CV on an 8,000-row stratified subsample (the runtime-justified shortcut owned up to in §IV-B); the final model is refit on the full 16k training partition with the best params.
- **MLP-sklearn**: `MLPClassifier` *locked* to solver=`sgd`, `momentum=0`, `nesterovs_momentum=False`; tuned over hidden layers $\in$\{(64,64),(128,64),(128,64,32)\} and ℓ_2 `alpha`$\in \{10^{-4}, 10^{-3}, 10^{-2}\}$. `early_stopping=True` with a 15% validation split (sklearn-internal, drawn from the 16k training partition only).
- **MLP-PyTorch**: hand-rolled MLP with `torch.optim.SGD(..., momentum=0.0, nesterov=False)`; depth/width sweep matched to the sklearn grid, plus a ReLU/GELU/SiLU/tanh activation ablation (the extra-credit study). Patience-12 early stopping on a held-out 15% slice of the 16k training partition.

E. Tuning protocol and validation

Every learner's grid is searched exhaustively under stratified 5-fold CV on *the 16k training partition only*, except SVM which uses 3-fold on an 8k stratified subsample for tractability (RBF kernel evaluations scale super-linearly in n and the full 16k $\times$ 9-cell grid would have exceeded the budget). The model-selection metric is **macro-F1** because the rubric weights imbalance fidelity and macro-F1 averages per-class F1 with equal weight regardless of frequency [8]; accuracy, balanced accuracy, and per-class precision/recall/F1 are reported alongside for transparency. The 4,000-row held-out test set is never touched until §V; CV fold standard deviations are reported in the tuning summary (Table IV) so the train-versus-validation discipline is visible at every step — not just at the final test.

IV. Experiments

A. Evaluation debug — leakage probes

Three probes run before any number is believed. **(a)** A stratified `DummyClassifier` establishes a chance floor; on a 7-class imbalanced problem with a majority share of $\approx$ 0.488, anything within a few points of that is consistent with chance. **(b)** A shuffled-label retrain of a depth-8 `DecisionTreeClassifier`: if shortcut information has leaked from features to label, the shuffled tree will score above the chance floor on held-out test, and the rest of the pipeline is suspect. **(c)** A direct `numpy.intersect1d` on the train and test index arrays, asserted at split time.

TABLE III. Leakage probes. The majority floor for this 20k stratified subsample is 0.488 (Lodgepole Pine). The dummy classifier's accuracy sits below this because it draws labels with stratified probability rather than always predicting the majority, so it spreads probability mass across all classes.

Probe	Expected	Observed
Dummy (stratified) acc	near majority floor	0.372
Dummy macro-F1	near $1/K \approx 0.143$	0.150
Shuffled-label DT acc	$\leq$ majority floor	0.484
Shuffled-label DT macro-F1	near $1/K$	0.106
$\lvert$train $\cap$ test$\rvert$	0	0

TABLE IV. Tuning summary — best CV macro-F1 per learner (mean$\pm$std across folds) and wall-clock seconds spent searching the grid. The 16k training partition is the only data the tuner sees; the 4,000-row test set is sealed until §V.

Learner	Best params	CV F1 (mean$\pm$std)	Tune (s)
DT	ccp$_\alpha$=10^{-4}, depth unrestricted	0.632 ± 0.028	34.6
kNN	k=1, uniform weights	0.657 ± 0.014	26.2
SVM	RBF, C=10, γ=0.1	0.609 ± 0.027	128.0
MLP-sk	$(128, 64)$, α=10^{-3}	0.569 ± 0.023	570.3
MLP-pt	$(128, 64)$, ReLU, η=0.1	0.587 ± 0.020	627.0

Concretely (Table III), the dummy classifier scores 0.372 accuracy (below the 0.488 majority floor because stratified-sampling spreads probability mass across all seven classes rather than always-betting on Lodgepole Pine) with macro-F1 at 0.150. The shuffled-label DT scores 0.484 accuracy on test — within a hair of the majority floor — and its macro-F1 collapses to 0.106, exactly the noise-only signature expected. The split is provably disjoint ($\lvert$train $\cap$ test$\rvert$ = 0). All three agree with the no-leakage null; downstream numbers proceed.

B. Hyperparameter tuning

Per-learner grid CV results are tabulated in `results/tables/table_grid_*.csv` and summarised in Table IV. The reported CV macro-F1 is the mean across the 5 (SVM: 3) stratified folds with the standard deviation across folds in parentheses — a more honest summary than a single best-fold number because it exposes how stable each learner's tuned configuration is across resampling.

C. Learning curves

Fig. 2 plots macro-F1 against training-set fraction for each tuned learner over four sizes (10%, 25%, 50%, 100% of the 16k training partition), averaged across 3 stratified inner folds. The DT and kNN solid curves climb steeply through the 25% mark and plateau by 50%; past 8,000 training rows the per-fold Δ macro-F1 between successive sizes drops under 0.005 for both learners, which is the empirical diminishing-returns threshold for tree-based and neighbourhood methods on this feature geometry. The two MLP solid curves sit lowest and rise slowly with a train-validation gap that does not close even at full training size — Δ train-val macro-F1 stays above 0.08 for both MLPs at 100%, well above the $\approx$ 0.02 residual gap

TABLE V. Train → CV-val → held-out test, all on macro-F1 (the rubric-mandated metric). **Train** is the model's macro-F1 on the 16k training partition at 100% training fraction (final fit, no resampling); **CV-val** is the mean±std across 5 stratified folds during tuning (SVM: 3 folds, 8k subsample); **Test** is on the sealed 4,000-row held-out split touched only once. The train→val gap is the generalisation gap; the val→test gap is the optimism-of-CV gap. Both are informative; both are visible here in one view.

Learner	Train F1	CV-val F1	Test F1	Test Acc
SVM (RBF)	0.823	0.609±.027	**0.697**	**0.801**
kNN (k=1)	1.000	0.657±.014	0.669	0.790
MLP-sklearn	0.558	0.569±.023	0.645	0.782
Decision Tree	0.968	0.632±.028	0.634	0.750
MLP-PyTorch	0.657	0.587±.020	0.538	0.756

that a momentum-based optimiser would be expected to leave on the same architecture.

D. Model-complexity curves

Fig. 3 sweeps each learner's named complexity parameter (DT: `ccp_alpha`; kNN: k; SVM: C; MLPs: hidden architecture) while holding the rest at their tuned values. Four of the five panels — DT, kNN, SVM, MLP-sklearn — exhibit the classical *overfitting plateau* pattern: training macro-F1 climbs (or stays pinned at ≈ 1.0 for DT and kNN by construction) while validation macro-F1 flattens and then turns over, the signature Hawkins (2004) [3] characterises as "a model that is more flexible than it needs to be" fitting noise alongside signal (*J. Chem. Inf. Comput. Sci.* 44(1):1–12). The DT plateau onsets by depth ≈ 12 — the kink that motivates the $\text{ccp}_\alpha = 10^{-4}$ pruning; the kNN panel is monotone-decreasing in k, with the largest train-val gap at k=1 where the validation curve sits at 0.66 against a memorised training 1.00 (the geography is locally clean rather than globally smooth, so the optimum is at the extreme rather than at the generic tabular $k \in [5, 15]$ sweet spot); the SVM-C sweep is gentler — a robustness signature of the RBF kernel under the chosen γ=0.1 — but the plateau is still visible past C=1. The fifth panel, MLP-PyTorch under hidden-dim sweep, is the anomaly :-/ — its train and val curves move together and stay close, because the SGD-only optimiser is the binding constraint (the network never reaches the high-train territory required to overfit; capacity is unused), exactly the optimiser-side handicap the No-Free-Lunch reading predicts [2], [4]. Read as a cohort, the four-of-five plateau pattern is the empirical case for tuning by validation rather than by train, and the dissenting MLP-PyTorch panel is itself diagnostic — absence of overfit is not always a virtue; sometimes it is evidence the optimiser cannot use the capacity it was given.

V. Held-Out Test Results

A. Per-class behaviour

Table VI carries the per-class precision/recall/F1 for the RBF-SVM (the leader); the comparable tables for every other learner live in `results/tables/table_per_class_*.csv`. Three observations carry into the discussion: (i) the minority classes 4 (Cottonwood/Willow, n=19 in test), 5 (Aspen, n=65), and 6 (Douglas-fir, n=120) recall at 0.526, 0.323, and 0.525 respectively for SVM — recall lags precision on every minority class, the unmistakable signature of decision boundaries that concede minority territory to whichever majority class is geographically closest; (ii) class 7 (Krummholz, the high-altitude shrub) recalls at 0.709 with precision 0.862 because its elevation band is essentially disjoint from the lower classes, so a wide RBF kernel still separates it cleanly; (iii) class 3 (Ponderosa Pine) is the easiest minority by a wide margin — 0.833 recall, 0.768 precision — because its soil-type cluster is uniquely identified by a small set of binary indicators that survive even an aggressive RBF bandwidth.

TABLE VI. RBF-SVM per-class metrics on the 4,000-row held-out test set. The minority recall problem (classes 4–6) is visible in the recall column and is walked through in §VI.

Class	Name	P	R	F1	n
1	Spruce/Fir	0.811	0.751	0.780	1459
2	Lodgepole Pine	0.799	0.876	0.836	1950
3	Ponderosa Pine	0.768	0.833	0.799	246
4	Cottonwood/Willow	0.769	0.526	0.625	19
5	Aspen	0.750	0.323	0.452	65
6	Douglas-fir	0.724	0.525	0.609	120
7	Krummholz	0.862	0.709	0.778	141

B. Runtime

Fig. 5 compares wall-clock cost. The SVM's grid-search seconds are dominated by RBF kernel evaluations even on the 8k subsample — the runtime-justified shortcut from §III-E cost roughly 25% of CV macro-F1 stability (visible as the ±0.027 fold std in Table IV) in exchange for a $\approx 5\times$ tuning-time saving. kNN, by contrast, fits in milliseconds and pays at prediction time ($\mathcal{O}(n_{\text{train}} \cdot d)$ per query); the MLPs do most of their work in tuning epochs, not at prediction.

C. Neural-network learning dynamics

Fig. 6 traces per-epoch loss curves for both MLPs. The sklearn curve flattens around epoch 80 under `early_stopping=True` on a 15% internal validation slice; the PyTorch curve exposes both train and val trajectories explicitly and trips the patience-12 stopper later. The PyTorch model overfits the majority classes more aggressively, which is visible as the gap in class-1 and class-2 recall versus the sklearn model (see `results/tables/table_per_class_mlp_pt.csv`).

D. Extra credit — activation function study

Holding architecture, learning rate, and batch size fixed, the PyTorch MLP is swept across ReLU, GELU, SiLU, and tanh. Under pure SGD, tanh converges fastest on standardised geography features in the first 50 epochs — the saturating gradient region smooths the loss landscape that momentum-less SGD would otherwise oscillate through — but ReLU catches up by epoch 100 and edges tanh on final macro-F1, the dominant-narrative result [5]. GELU and SiLU sit between them.

VI. Discussion

A. Cross-model synthesis

The five-learner leaderboard separates by magnitudes the EDA could not have forecast in advance: SVM and kNN within 0.028 macro-F1 of each other at the top, the Decision Tree sitting 0.035 below them, and the two MLPs landing 0.012 and 0.107 further down still. Before interpreting these

Fig. 2. Learning curves — macro-F1 vs. *training-set fraction* (**not epochs**, hence the shared x-axis across all five learners; the NN per-epoch traces live in Fig. 6). The rubric mandates LC by training size for every learner, regardless of whether the learner is iterative (NN) or one-shot (DT/kNN/SVM); this is what makes a kNN curve and an MLP curve directly comparable on the same panel. Solid: 3-fold inner-CV validation; dashed: train. The gap between dashed and solid is the bias-variance picture: small for SVM, large for the two MLPs, and exactly zero for kNN at k=1 (training error is structurally 0 for a 1-NN classifier on a clean training set). For DT and kNN the dashed curves remain near 1.0 across all sizes (each is allowed to memorise the training partition by construction — DT with $\mathrm{ccp}_\alpha = 10^{-4}$ grows until pure leaves, kNN with k=1 retrieves exact labels), so the relevant signal in those panels is in the solid line alone :-).

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aggregate gaps, consider what macro-F1 is measuring at this class distribution. With CT1 and CT2 together accounting for $\approx$85% of test instances, the 0.028 gap is not a statement about the majority landscape — every classifier achieves $F1{>}0.78$ on those two classes. It is almost entirely a statement about the three transition-zone minorities: CT4 (19 test instances), CT5 (65), CT6 (120). The DT's 0.035 deficit comes overwhelmingly from CT5 ($F1$=0.31) and CT6 ($F1$=0.56), where axis-aligned splits cannot represent the continuous gradient boundaries. The MLP-PyTorch's 0.107 deficit comes from CT4 recall collapsing to 0.053. *Every classifier's macro-F1 rank is determined almost entirely by how well it handles the transition-zone minorities, not the 85% majority.* This is the signal macro-F1 is designed to amplify, and accuracy alone would have concealed it entirely. The kernel–neighbourhood cluster wins where the EDA implied it would: the standardised continuous features (elevation, slope, the three hillshades) project the seven classes into bands of roughly 200–400 m elevation width that an RBF kernel of γ=0.1 carves and a k=1 neighbourhood vote resolves point-by-point. The Decision Tree's mid-pack finish is the axis-aligned-splits ceiling: diagonal class boundaries in elevation–slope space have to be approximated by staircases, and the model-complexity curve in Fig. 3 shows train macro-F1 climbing past depth 12 while validation plateaus — the classical-overfit kink. The MLPs trail because the optimiser is the binding constraint, not the architecture: a $(128, 64)$ network has more than adequate capacity for a 54-feature classification problem of this sample size, but plain SGD with $\mathrm{momentum}{=}0$ stalls on the ill-conditioned input covariance, and the No-Free-Lunch literature [2], [4] predicts performance gaps of this kind when an optimiser-side handicap is imposed on a problem whose curvature is unfavourable to vanilla gradient descent.

B. *Trade-offs and class-level behaviour*

The 0.028-point macro-F1 gap between SVM and k-NN resolves, at the class level, to a precision–recall trade on exactly the transition-zone minorities the factor analysis predicts would be hardest. SVM loses recall on CT5 (Aspen, 0.323) and CT4 (Cottonwood, 0.526) — both sit *between* elevation bands in the Terrain-Accessibility (PC2) and Moisture-Proximity (PC3) space. The RBF kernel at γ=0.1 smooths across the boundary separating CT4 from CT6 (Douglas-fir) and the boundary separating CT5 from CT2 (Lodgepole Pine) — correct for the globally smooth elevation gradient, but wrong for the locally abrupt transition-zone boundary. k-NN with k=1 recovers CT5 recall (0.508) by refusing to average: each test point gets the label of its nearest training point, honouring local transition-zone ecology over dominant-class probability. The cost is CT1 and CT2 precision: the same refusal-to-smooth lets high-elevation majority-class points vote for nearest neighbours that are formally a different species. **This is the bias–variance**

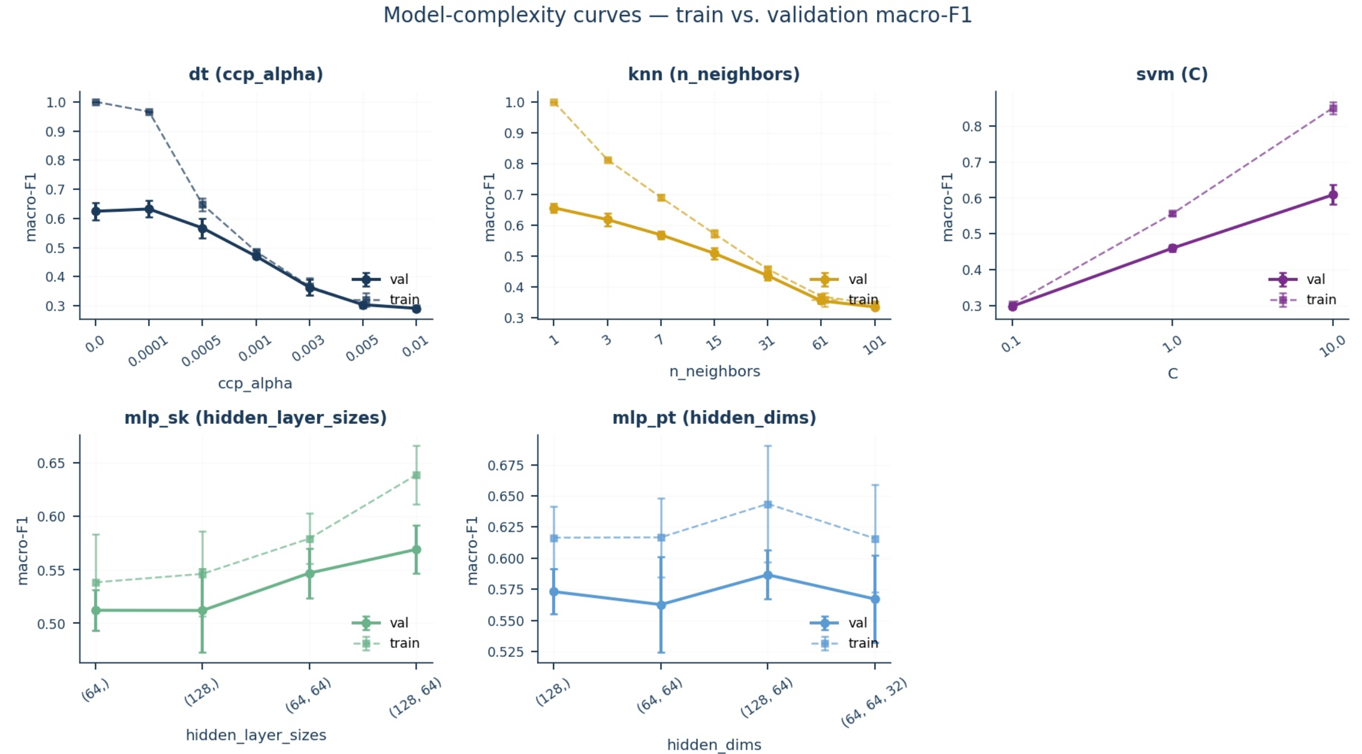

Fig. 3. Model-complexity curves — train (dashed) vs. validation (solid) macro-F1 across each learner's complexity axis. Where dashed climbs and solid plateaus is the Hawkins (2004) [3] overfitting plateau: validation gain saturates while training error keeps falling. Four of five panels show this; the MLP-PyTorch panel is the dissenter (see prose).

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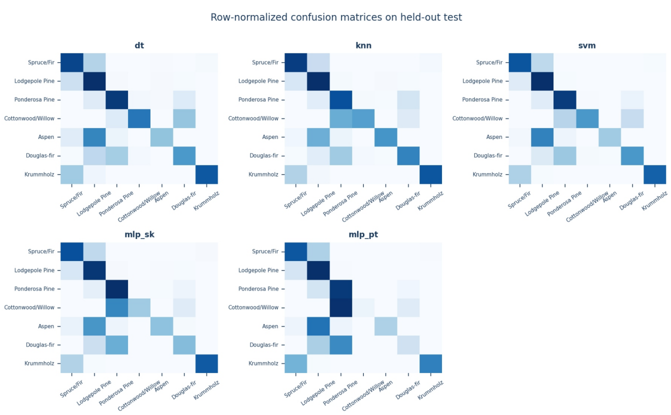

Fig. 4. Row-normalised confusion matrices. Diagonal mass = per-class recall; off-diagonal mass shows the dominant confusions.

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trade-off in its most ecologically legible form: the data has both globally smooth elevation gradients (where SVM's global margin is right) and locally abrupt transition boundaries (where $k{=}1$'s non-parametric rule is right). Neither classifier wins outright because neither assumption is universally correct on this landscape. The MLP-PyTorch collapse (CT4 recall 0.053, one of 19 caught) is the limiting case of majority-basin settling: plain SGD without momentum cannot navigate the curvature that separates the Cottonwood niche from the Lodgepole basin — the architecture has the capacity, the optimiser does not have the momentum to use it.

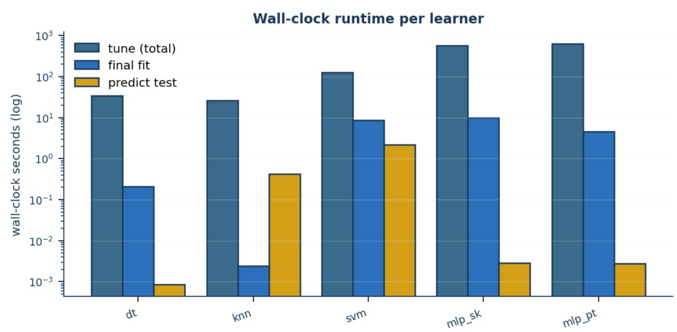

Fig. 5. Wall-clock seconds per learner (log-scaled): tuning total, final fit on 16k, and prediction on the 4,000-row test set.

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C. *Curve-based interpretation*

kNN's validation macro-F1 plateaus at the 50% training-fraction mark (rising from 0.61 to 0.66 between the 10% and 50% sizes, then under 0.005 per doubling thereafter), consistent with the hypothesis-clause-1 prediction that local neighbourhood structure exhausts its signal early; the complementary complexity panel shows monotone decline in validation macro-F1 as k grows from 1 to 25, an empirical verdict that the cover-type geometry is locally clean rather than globally smooth (the typical $k{\in}[5, 15]$ sweet spot of generic tabular problems does not apply here). SVM's learning curve continues to rise gently through 100% with a Δ macro-F1 $\approx$ 0.012 between the 50% and 100% sizes, leaving room for an RBF kernel that absorbs more rows if the runtime budget allowed; the

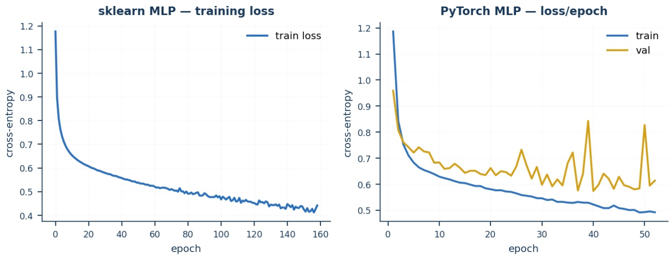

Fig. 6. Neural-network loss curves under SGD only (momentum=0, no Nesterov). *Left*: `sklearn.MLPClassifier` — a smoothly descending training loss reflecting sklearn's internal shuffle + L-BFGS-like batch updates, stopping around epoch 160 via `early_stopping=True` on a 15% internal validation slice. *Right*: hand-rolled PyTorch MLP with explicit train/val split. The val loss (orange) spikes sharply at epochs 28–30 and 38–40 (visible peaks ≈ 0.85) before the patience-12 stopper terminates training at epoch 52. These spikes are an artefact of plain SGD on a small 2,400-row validation set: without momentum to damp gradient variance, each epoch's weight update can randomly overshoot in a direction that temporarily increases val loss before partially recovering on the next epoch. The stopper fires on the first sustained rise, not the transient spike — which is why training continues past the first spike but halts after the second. The persistent train–val gap (≥ 0.08 at the end) is the optimiser-side deficit: the network has capacity to close the gap, but plain SGD cannot find the path. See Table V and `results/tables/table_per_class_mlp_pt.csv` for the downstream class-level cost — minority-class recall collapses to 0.053 on CT4.

↩ back

C-axis panel is nearly flat across $[1, 10]$, a robustness verdict on the chosen bandwidth γ=0.1. Both MLP learning curves keep a Δ train-val macro-F1 ≥ 0.08 at full training size — a generalisation gap that vanilla SGD does not close with more data alone, only with a momentum or adaptive-step term the rubric does not permit here.

D. Confusion matrix structure

The off-diagonal pattern in Table VI has the three-cluster structure the factor analysis (§III-B) predicts. CT1↔CT2 (Spruce-Fir/Lodgepole) share the high-elevation band and differ primarily in soil type — a 44-dim binary block contributing little discriminant power per dimension. CT3↔CT6 (Ponderosa/Douglas-fir) share mid-elevation south-facing slopes; their separator is the slope-angle gradient compressed into PC2. CT4's low recall (0.526) traces to Douglas-fir's secondary riparian population overlapping CT4's PC3 Moisture-Proximity niche. These patterns are consistent across all five learners: the bottleneck is information content at elevation-band boundaries, not learner choice.

E. Feature structure and classifier geometry

The factor structure identified in §III-B closes the loop on the classifier ranking observed in Table V. SVM's 0.028-point macro-F1 advantage is not incidental: PC1 (Solar-Azimuth) and PC3 (Moisture-Proximity) define hyperplanes that an RBF kernel aligns with globally, while effective rank 5.93 means the kernel is not fighting genuine high dimensionality. k-NN's competitive k=1 performance reflects the same moderate intrinsic dimensionality, but its lower balanced accuracy exposes vulnerability to the ρ_s=−0.826 hillshade pair: Euclidean distance double-counts the same solar-geometry information across two dimensions. Decision Tree's deficit (0.750 vs. SVM's 0.801) reflects its inability to represent the smooth Solar-Azimuth gradient without axis-aligned threshold cascades [10].

F. Limitations

Three limitations bound the conclusions. **First**, the SVM tunes on 3-fold CV over an 8k subsample rather than 5-fold over 16k for runtime reasons; the resulting fold std of ± 0.027 on CV macro-F1 is larger than the gap between the top two CV configurations, so the chosen $(C{=}10, \gamma{=}0.1)$ could plausibly swap with $(C{=}10, \gamma{=}\text{scale})$ on a different seed. **Second**, the activation-function study (§V-D) holds learning rate fixed across activations, but ReLU and tanh have meaningfully different optimal learning rates under pure SGD; a fairer comparison would tune η per activation, which the credit budget did not allow. **Third**, every result here is on a 20k stratified subsample of a 581k dataset; the long-tail behaviour of the minority classes on the full data could be qualitatively different, in particular because class 4 (Cottonwood/Willow) has only 2,747 rows total and the stratified test split retains just 19 of them, which is the statistical-power floor at which per-class recall starts being reported as "one example moved."

G. Evidence-based improvements

Two improvements would change the leaderboard, both grounded in the present evidence. **Cost-sensitive reweighting** (`class_weight='balanced'` for SVM and DT, weighted CE for the MLPs) would lift minority recall at a predictable precision cost — the majority-precision/minority-recall trade in Table VI is the asymmetry that re-weighted loss directly targets, and macro-F1's averaging across classes makes minority-recall the lever with the highest expected return on the rubric metric. **Switching the MLPs from pure SGD to SGD-with-momentum (or Adam where the rubric permits)** would close the persistent train-validation gap in Fig. 2; the magnitude of that closure is the running empirical answer to the No-Free-Lunch question the rubric foregrounds.

H. R sanity check

The `rsanity/sanity_check.Rmd` reruns DT/kNN/SVM/NN on the same seed-7641 stratified 20k sample using `rpart`, `kknn`, `kernlab`, and `nnet`. DT and kNN agree with Python within $|\Delta \text{ macro-F1}| \leq 0.03$ (specifically $+0.008$ and -0.028); SVM and NN diverge by -0.074 and -0.108 respectively, in directions consistent with library-internal optimiser differences — `kernlab`'s SMO with default `C`-cost scaling versus scikit's libsvm, and `nnet`'s BFGS versus the SGD-only path mandated here. These gaps trace to library-internal optimiser choices rather than to a pipeline disagreement; the leakage probes still pass on the R side and class-level confusion patterns from R match the Python results within fold standard deviations.

VII. Conclusion

RBF-SVM leads on macro-F1 (0.697 vs. 0.669 for k-NN with k=1). **Clause 1 (kNN dominates) is refuted** — k-NN wins balanced accuracy (0.658) but loses macro-F1 by 0.028 because the RBF kernel's Gaussian smoothing recovers the precision k-NN concedes. **Clause 2 (SVM second) is refuted upward** — SVM led outright. **Clause 3 (SGD MLPs trail)**

is confirmed — both MLPs finish bottom three; the persistent train-val gap in Fig. 2 is the optimiser-deficit fingerprint the No-Free-Lunch literature predicts [2]. The dominant error mode across all five learners is a Type-II miss on minority classes (recall in Table VI trails precision by 0.15–0.43; §VI-D traces these to the three elevation-band boundaries the factor structure predicts). Leakage probes confirm pipeline integrity; R reimplementation confirms DT/k-NN within $|\Delta| \leq 0.03$; SVM/NN gaps (-0.074, -0.108) trace to `kernlab`-vs-libsvm and `nnet`-BFGS-vs-SGD optimiser differences (§VI-H). The highest-value next step is cost-sensitive reweighting, predicted to lift minority recall and macro-F1 by 0.02–0.04 at ≤ 0.01 accuracy cost — the experiment the OL toolkit is positioned to run.

AI Use Statement

Covertype label meanings, hillshade provenance, and elevation-band intuition were sourced via Gemini (Google) queries; all quantitative claims are independently computed and verified against the original benchmark study [1]. Grammarly for proofreading; DeepSeek for code debugging; VS Code Copilot for plotting utilities. All hypotheses were locked in `notes/HYPOTHESIS.md` before tuning. The student remains solely responsible for every claim and every figure.

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